Mathematical Modeling of Influenza Using SIR and SIRS Models

Final project Report

CSCI 2897 : Calculating Biological Quantities

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The outbreak of the influenza virus has been a serious issue in public health, as it keeps coming back. These types of diseases are easier to handle with mathematical modeling to understand how it spread and how interventions may control outbreaks. The most common models are SIR and SIRS, which divide a population into susceptible, infected, and recovered groups. The SIR model assumes that recovered individuals gain permanent immunity, while the SIRS model allows immunity to fade, permitting reinfection.

The main goal of this project is to understand how different factors affect the spread of a disease using the SIR and SIRS models. Specifically, this study examines the effects of the infection rate (β), recovery rate (γ), immunity loss rate (ω), and vaccination rate (v) on the number of infected individuals over time. By changing these values and comparing the results, this project shows how changes in transmission, immunity, and vaccination influence outbreak severity, duration, and long-term behavior. These results help explain how public health actions, like vaccination, can change the outcome of an outbreak.

The SIR and SIRS models were implemented using a system of ordinary differential equations.
The equations for the SIR and SIRS models are
SIR
$dS \div dt = -\beta SI - vS,$
$dI \div dt = \beta SI - \gamma I,$
$dR \div dt = \gamma I + vS$

SIRS
$dS \div dt = -\beta SI + \omega R - vS,$
$dI \div dt = \beta SI - \gamma I,$
$dR \div dt = \gamma I - \omega R + vS$

Where S(t) means the number of susceptible individuals, I(t) means the number of infected individuals, and R(t) means the number of recovered individuals. And each parameter, like β the infection rate, meaning how fast the disease spreads; γ the recovery rate, meaning how fast infected people recover; v the vaccination rate, meaning how fast susceptible people are being vaccinated and moved into the recovered group; and ω the immunity loss rate, meaning how fast recovered people become susceptible again.

And we used the total population of 10,000 individuals with initial conditions of 9,990 susceptible, 10 infected, and 0 recovered. For my model simulations were performed over a 200 day period. A baseline recovery rate of $\gamma = 0.2$ was used, corresponding to an average infectious period of 5 days.

To understand the effect of each parameter in depth, different values, low and high, were tested. The value of parameters that are tested in this model are

β	1×10^{-4}	5×10^{-4}
ω	0.02 (slow loss)	0.5 (fast loss)
v	0 (no vaccination)	0.5

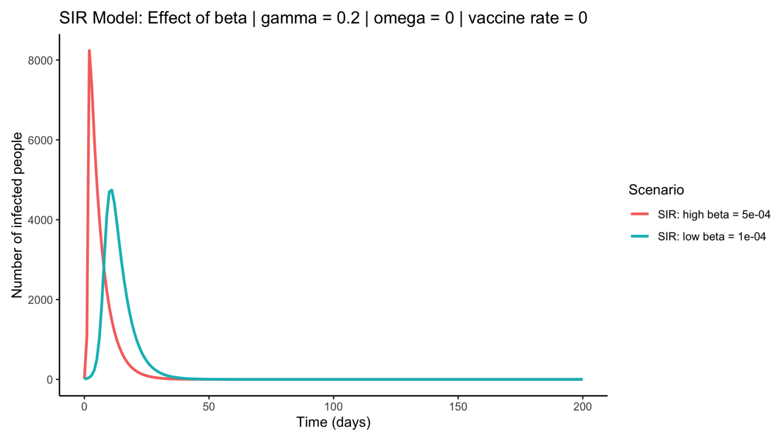
And to see the effect of these parameters, time series simulation methods were used.

The simulation results show that increasing the infection rate leads to a much faster and higher epidemic peak in both SIR and SIRS models. When β was high, infections rose sharply and peaked quickly above 8,000 individuals. Whereas, lower β produced a slower outbreak with a lower maximum number of infected individuals and a delayed peak.

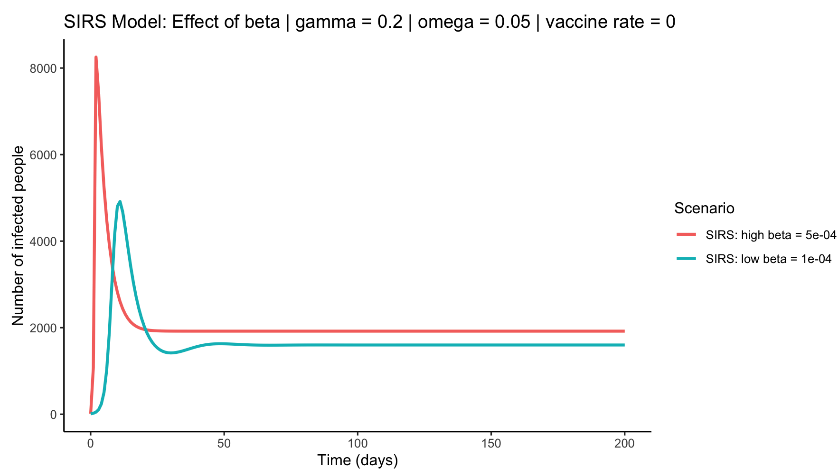
In the SIR models without immunity loss, all infection curves eventually declined to zero, indicating that the disease naturally dies out once immunity accumulates in the population. However, the SIRS model produced a markedly different long-term behavior. When immunity loss ω was included, infections did not disappear completely. Instead, the system reached a stable endemic equilibrium where hundreds to thousands of individuals remained infected indefinitely. Higher ω values produced higher long-term infection plateaus.

Vaccination produced the most dramatic effect across all simulations. When v = 0.5, the infection peaks were reduced by more than 80–90% in both SIR and SIRS models. In many vaccinated scenarios, infections remained near zero for the entire simulation. In the SIRS model, vaccination also significantly suppressed the endemic equilibrium level, demonstrating that vaccination compensates for immunity loss and prevents long-term persistence.

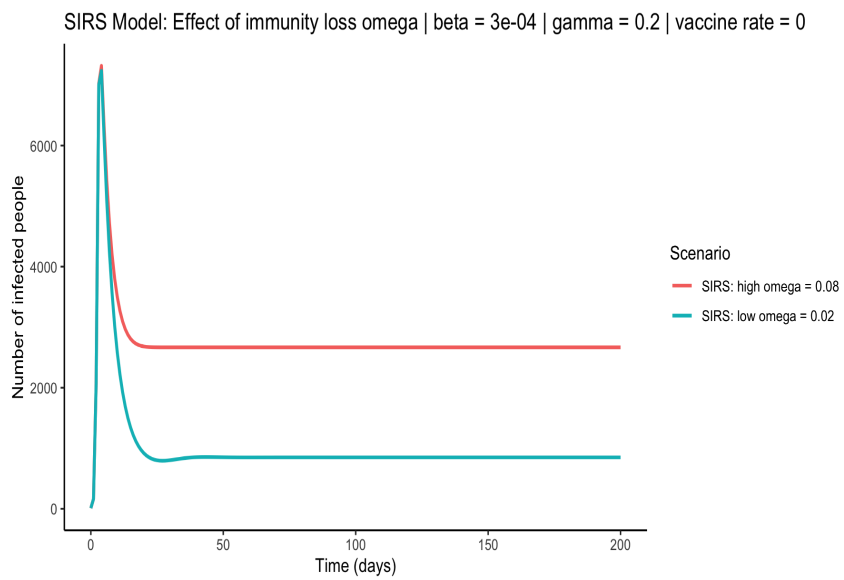
The following figures present the simulation results generated in this project to show how changes in infection rate, immunity loss, and vaccination affect disease dynamics.



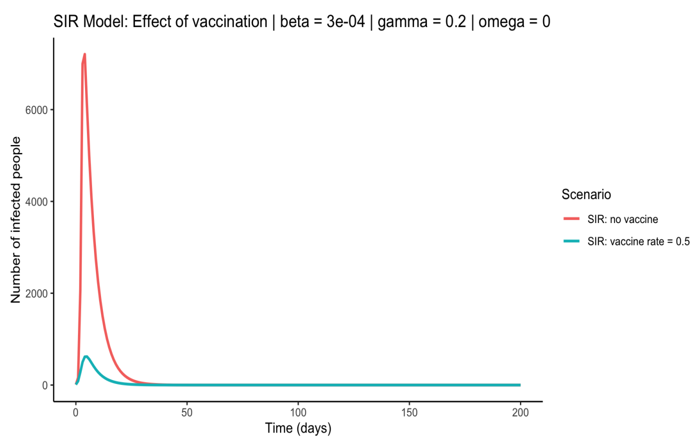
This graph illustrates the effect of infection rate β, where a higher infection rate produces a faster and larger outbreak peak, while a lower infection rate delays the peak and reduces outbreak size. In both cases, infections eventually decline to zero due to permanent immunity



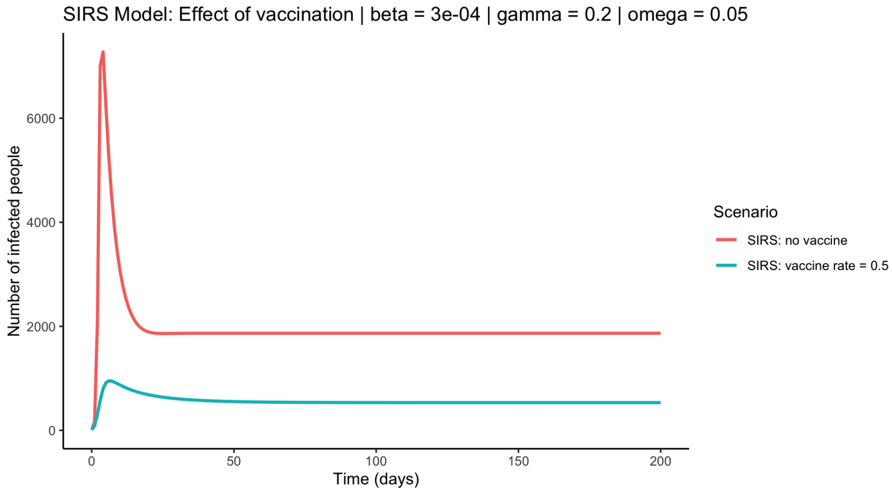
This graph illustrates the effect of infection rate where a higher infection rate leads to a faster outbreak, a larger peak in infections, and a higher long term endemic level. A lower infection rate results in slower growth, a smaller peak, and reduced endemic infection levels.



This graph illustrates the effect of immunity loss rate, where a higher loss rate leads to a larger long term endemic infection level, while a lower immunity loss rate reduces reinfection and lowers sustained transmission



This graph illustrates the effect of vaccination in the SIR model, where it shows that without vaccination, the epidemic reaches a large peak. With vaccination (v=0.5), the peak number of infected individuals is drastically reduced and the outbreak dies out much faster.



This graph illustrates the effect of vaccination in the SIRS model, where it shows that without vaccination, the infection reaches a high peak and settles into a large endemic level. With vaccination (v=0), both the outbreak peak and long-term endemic infection level are substantially reduced.

These results show that the infection rate β plays the biggest role in how quickly and how seriously an outbreak spreads. Higher β values mean increased contact rates or higher transmission probability and lead to explosive outbreaks with very high peaks. When β is low, the disease spreads more slowly, the number of infected people is smaller, and the outbreak is easier for hospitals and healthcare systems to manage.

The immunity loss parameter ω introduces a fundamentally different disease behavior. In the SIR model, immunity is permanent, so outbreaks always fade out. However, the SIRS model allows reinfection, which produces sustained disease circulation. This behavior closely resembles real world seasonal diseases such as influenza, where immunity is not permanent. Increasing ω directly increases the long term endemic infection level.

In every tested simulation, vaccination proved to be the most successful method of controlling the illness. The number of infected individuals significantly decreased, the outbreak occurred later, and in certain situations, it was totally avoided when the vaccination rate was high. Vaccination helped maintain low infection levels and prevented the disease from spreading for extended periods of time, even when immunity gradually declined. These findings demonstrate how crucial immunization is for stopping both transient outbreaks and chronic illness.

Bibliography

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